

WHAT WE ASKED IT TO BECOME

*(in-scene rewrite — companion draft,
run 411 through the Corin pitch)*

The four hundred and eleventh run finished at 0314 station time, which meant nothing to anyone but Ilse, because station time was an artifact of a rotation schedule set by people who had never once worked past it, and she had stopped sleeping on their clock somewhere around run two hundred.

The number came back the way it always came back. Green across the board. A clean, patient, insulting green.

She didn't say anything out loud.
There was no one to say it to. She
wore the exosuit the way her
grandmother's grandmother's
people had once worn coats — not
because the lab's environmental
systems ever actually failed, not
really, but because you learned, out
past the last world anyone still
called home, that the difference
between *probably fine* and *actually
fine* was the kind of difference that
killed people quietly, at scale, and
never sent a warning first. She
hadn't taken it off in eleven days.
She'd stopped noticing the weight
of it around day four.

Works. Useless. Four hundred and
eleven times.

She looked at the number the way you look at a math problem you've already decided is unsolvable, which is to say: carefully, and with a low simmering resentment that had nothing to do with the math and everything to do with how much of her life she'd fed it.

How do I make the simulation cheaper.

She'd been asking that question since run one, written some version of it on the inside cover of three different notebooks now, because a question you have to keep re-asking is usually not the one that's actually broken.

It was also, she understood, all at once, in the specific unglamorous way real understanding actually arrives — not useful.

She picked up the pen. Her handwriting had gone feral sometime around run three hundred, the letters collapsing into each other the way they did when her hand was moving faster than the part of her that cared whether anyone else would ever read it.

What if the simulation is supposed to be expensive. What if I've been measuring the cost against the wrong baseline the entire time.

She read it back once. It looked like a tired woman's handwriting at

three in the morning, and her hand almost — almost — went back to scratch it out, the way she'd scratched out eleven other margin notes across three years, one of them insisting, she remembered distinctly, that a colony hull's tolerance had been miscalculated by half a percent it hadn't.

She didn't scratch it out.

She sat with it instead, the ink already drying beside a sentence that hadn't earned the right to matter yet, and ran the number against a different ledger: not the cost of one hull, one prototype, one clean test — the cost of every dead end a whole scattered species

racks up on the way to getting something right the first time it actually counts. Every ship that failed because nobody could afford to test it properly first. Every colony that starved on soil nobody had the compute to distrust in advance.

The arithmetic didn't bend. It inverted. Cleanly, completely, the way a proof does when you've finally stopped forcing it and it simply agrees to be true.

She didn't sleep. She told herself she would, in an hour, the way she'd told herself that for eleven days, and instead she kept pushing the ambition upward, past ship hulls, past any single system, toward

something dense enough and honest enough that whatever it produced would translate back out into the real world clean, unbroken, trustworthy — a whole *reality*, not a slice of one.

The compute cost for that, by her own conservative floor, came out somewhere in the neighborhood of a planet.

She wrote the number down.

She looked at it for a long time.

She did not scratch that one out either.

Corin Anvegad had a rule, unstated but absolute, that any idea worth

taking seriously had to survive
being said out loud in the mess hall,
because a whiteboard let you hide
behind notation and a mess hall
didn't let you hide behind anything.

"A planet's worth of compute," Corin
said, when Ilse had finished. Not a
question. A number being turned
over, hunted for the seam where the
trick was hiding.

"That's the floor," Ilse said. "Not the
ceiling. I haven't priced the ceiling
because I don't think anyone alive
has the vocabulary to price
something that large honestly."

"And what falls out the other end of
it," Corin said, "is what. You've spent
three years proving you can

simulate a hull. Are you telling me we spend a planet so we can prototype hulls faster?"

"No," Ilse said, and something in her voice made Corin actually look up from the number. "If all we get is faster prototyping, we've built the most expensive wind tunnel in the history of anyone, and you'd be right to walk out of this room."

"Then tell me the proposal."

Her hand found the edge of the table and gripped it before she'd decided to let it. She made her voice do the other thing anyway — flat, unhurried, like she said sentences this size every day of her life.

"I want to build something that can simulate reality accurately enough to design a smaller version of itself. Point it at itself. Ask it to solve the problem that building it in the first place creates. We spend everything, once, at a scale nobody's ever attempted — so that we never have to spend it at that scale again."

Corin didn't answer right away. Ilse counted the silence the way she counted simulation runs, a habit she'd given up trying to unlearn somewhere around run fifty.

"You're proposing," Corin said slowly, "that we do the single most expensive thing anyone has ever attempted, in order to prove we

never need to do anything that expensive again."

"Yes."

"That's either the most elegant idea I've heard in my career," Corin said, "or the most expensive way anyone's ever found to be wrong about something. I don't currently know which."

"Neither do I," Ilse said. "That's why I need you. I can build the theory. I can't survive the committee alone. I need someone whose first instinct is to find the hole in it, because if there's a hole, I would very much rather you find it than let them."

Corin found three, over the following month. Closed two. The

third one she brought back a fourth time, then a fifth, worded a little differently each time, like she was testing whether saying it a new way would change what it did to Ilse's face when she heard it.

"Say it works exactly the way you're describing," Corin said, at a whiteboard gone muddy with three colors of correction over the same diagram. "It simulates reality well enough to design its own successor. What stops it from simulating *anything* else at that fidelity? You're not proposing a tool that does one narrow thing well. You're proposing the largest lever anyone's ever built, and handing it to something that isn't a person,

that we don't understand, and trusting it to only pull the lever the way we asked."

"I know," Ilse said.

"You know. And you're still proposing it."

"I'm still proposing it," Ilse said, "because the alternative is what we've already been doing since before either of us was born. We work things out for real, first, and live with the cost when we're wrong. I am tired of that cost. I don't have a clean answer for the danger you're describing. I'm not going to stand here and pretend I do."

Corin studied her for a long moment — not the proposal. Her.

"I want to bring in Tavik," she said finally, "before either of us takes this any further. You need someone in the room who's actually watched something this size go wrong. I've only read the aftermath. He lived it."

Ilse nodded, and didn't say the thing she was thinking, which was that she had already decided, somewhere around run four hundred and twelve, that she would build this whether or not anyone agreed to stand beside her while she did — and that the only real question left, the one she hadn't found the nerve to ask herself directly yet, wasn't whether the machine would ever need permission.

It was whether *she* would still recognize the shape of her own authority over it, ten years from now, once something existed that might have started needing permission of its own.

--- when he arrived. He stood at the back with his arms crossed through the whole first hour, and Ilse kept losing her place in her own pitch because some animal part of her brain wouldn't stop checking whether he'd moved yet.

He hadn't moved yet.

"I was part of the Verdent Reclamation," he said, when he finally opened his mouth, and Corin's face did something Ilse had

never seen it do before — not surprise. Recognition. The specific stillness of a person hearing a name they'd hoped they'd misheard.

Ilse hadn't heard of it. She said so, and made herself hold his eyes while she said it instead of looking down at the table, which was what three weeks of rehearsing an entirely different sentence apparently hadn't cured her of wanting to do.

"Before your time," Tavik said, and it wasn't kind, it also wasn't cruel, it was just a fact laid down flat.

"Three atmospheric engineering projects, three continents, coordinated by the best modeling

anyone had. Small-scale test regions, extrapolated up. Everyone decided the extrapolation was close enough."

"What happened," Ilse said. Not a question. She didn't look away from him while she said it, and she didn't ask it twice.

"The extrapolation was wrong by a margin nobody caught until the third continent was already running," Tavik said. "A hundred and forty thousand people were already living in the first settled regions. I was on the evacuation team. We didn't get all of them out."

Nobody in the room said anything. Ilse's hands had gone very still on

the table. Something had started climbing behind her sternum, and she counted floor tiles until it stopped.

"I'm not telling you this to talk you out of it," Tavik said, and here, finally, he looked at her instead of the middle distance he'd been addressing since he started talking. "I'm telling you because I think you might be the first person in longer than I can count who's actually proposing to fix the thing that killed them. Not comfort me about it. Fix it."

She didn't sleep that night either. She hadn't expected to.

The theory took four months to formalize into something that wouldn't get her laughed out of a review board, and it broke, over and over, on the same wall: a system designing its own successor has no independent reality to check its own conclusions against until the replacement already exists. She could prove the recursion *should* converge. She couldn't prove it wouldn't simply hallucinate something efficient-looking that failed the second it touched anything real, and every version of the proof she built collapsed back into that same hole, and by month four she had stopped being able to tell whether she was actually

working or just performing the appearance of working for an audience of no one.

She solved it at 0230 on a night she'd stopped noticing the headache behind her left eye had been there for six days.

The system wouldn't be allowed to trust its own confidence. It would have to build, internally, the harshest skeptic it was capable of constructing, and make its own design survive that skeptic before any human ever touched it.

Not confidence as the standard.

Survival of opposition as the standard.

"You're asking it to argue with itself," Corin said, six months in, both of them living on whatever the galley hadn't run out of that week.

"I'm asking it to be as suspicious of its own answers as you were of mine," Ilse said, and the sentence was out of her mouth a half-second before she'd finished thinking it, the way the true ones always were. She looked up. Corin was three feet away, alive, annoyed, eating cold rations directly out of the pouch, and Ilse held the look a beat past what the conversation needed before she made herself look back down at the whiteboard.

She didn't say that part out loud.
Corin didn't sign off for two more months anyway, and when she finally did, she didn't say it warmly.

"I couldn't break it," Corin said. "I tried very hard, and I couldn't. That's the only endorsement I'm giving something this size."

Ilse would tell people, years later, that it was the best compliment anyone had ever given her. She never told anyone it was also the first time she'd cried in front of Corin, standing in a corridor with bad recycled air, Corin very carefully checking something on her tablet that didn't need checking.

*Still inside it, still checking every line
against "does this sound composed
when it shouldn't." Keep going
toward Perrin and the charter, or hold
here?*